

PRAKASH S

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Summary

Software Development Engineer with **3 years of experience** building scalable backend systems and ML-powered features at Amazon and Bigthinx. Strong background in **distributed systems, backend development, and machine learning infrastructure**. Proven ability to own end-to-end features, reduce operational overhead, and deliver measurable business impact.

Experience

Software Development Engineer

Aug 2022 – Nov 2024

Amazon – Chennai, India

- Owned and delivered the **Incoming Call Blocking** feature for Alexa, reducing monthly user churn by **10%** across millions of devices.
- Designed and implemented a **malware scanning service** for file uploads, strengthening security within the Alexa LLM architecture.
- Built and launched an experimental **primer page** to optimize Alexa device setup using **A/B testing** and user acceptance metrics.
- Improved system reliability by identifying bottlenecks and reducing **operational load** through automation and refactoring.
- Collaborated with product, QA, and operations teams to take features from design to production.

Computer Vision Engineer

Jun 2019 – Jan 2020

Bigthinx – Bangalore, India

- Designed **process-based parallelism in Python**, reducing image processing server latency by **60%**.
- Implemented ML-based **object detection models in C++** and deployed them on edge devices.
- Researched and prototyped computer vision and deep learning algorithms for image recognition use cases.

Education

Bachelor of Engineering in Mechatronics

2015 – 2019

Anna University, Coimbatore, India

Technical Skills

Languages: Python, Java, C, C++, JavaScript, SQL

Frameworks & Libraries: Spring, TensorFlow, Keras, Flask, OpenCV, ReactJS, Node.js

Systems & Tools: AWS, Docker, MySQL, MongoDB, Distributed Systems, Data Structures & Algorithms

Machine Learning: Computer Vision, Model Deployment, Inference Optimization

Projects & Open Source Contributions

Keras Contribution

Contributed a dataset splitting utility to **Keras**, a deep learning framework built on TensorFlow and used by over **1M+ developers**.

Numba

Added support Numba's Jit compiled np.mean to handle scalar inputs

Cpython

Fixed memory leak bug in Generic-Alias type implementation in python core compiler

DeFIR – Deep Feature Image Retrieval

Built an image similarity search system using deep learning embeddings and nearest neighbor search for content-based image retrieval.

LLM Agent

Entity: The AI agent that bridges you and your OS—seamlessly execute commands and control apps on Windows, macOS, and Linux.

Pyqoi

Developed a Python library implementing the **QOI (Quite OK Image)** format, providing faster and more efficient lossless image compression.